	SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P)
Department	Applied Science
Branch	**Group A**: Civil, Mechanical, Electrical, Electronics & Instrumentation, Electronics & Communication, CSE (IOT) **Group B**: CSE, CSE (BC), CSE (AIADS), Information Technology, CSE (AIML), CSE(Cyber Security)
Session	Batch 2026-27 onwards

Category of Course	Course Title	Course Code	Credits - 3			Theory Paper
			L	T	P	
BSC	Applied Chemistry	CH26101	3	0	0	Max.Marks - 70 Min.Marks - 22 Duration - 3 Hrs.

Course Code	Course Title	Maximum Marks Allotted						Total Marks
		Theory Paper			Practical			
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	Assign/Quiz /Term paper	
CH26101	Applied Chemistry	70	20	10	--	--	--	100

Course Objectives	The main aim of Engineering Chemistry is to make Students familiar with basic concepts of Chemistry, the students face in industry and engineering field. With this background the Students will be able to explain Scientifically the various chemistry related problems in industry/engineering field.
Prerequisite Knowledge	Students who have completed 12th with Science stream or Chemistry of 12th standard or equivalent.
Course Description	This course provides fundamental and applied knowledge of Chemistry in areas such as water technology, fuels, lubricants, corrosion, energy storage systems, polymers, nanomaterials, and instrumental analytical techniques. It emphasizes industrial applications, modern materials, electrochemical systems, and analytical methods along with numerical problem-solving and practical understanding of chemical processes.
Course Outcomes	**Upon completion of the course, student will be able to (Please co-relate 05 Co's with 05 Unit)** 1. Differentiate hard & soft water, solve the related numerical on water treatment and have knowledge regarding its significance in industry and daily life. 2. Students will acquire knowledge of fuel classification, analysis, calorific value determination, and the properties and industrial applications of lubricants. 3. Students will gain knowledge of corrosion mechanisms, prevention strategies, electrochemical cells, and the construction, applications, and recycling of Li-ion batteries. 4. Students will acquire knowledge of the synthesis, properties, and applications of polymers, conducting polymers, photoactive materials, and nanomaterials. 5. Students will gain knowledge of the principles, instrumentation, and applications of electroanalytical and spectroscopic techniques used in chemical analysis.

SYLLABUS

Unit - I	**WATER TECHNOLOGY:** Sources, Availability, impurities in Water, Types of hardness, Units of hardness. Concentration expression: Normality, Molarity, Molality. Water analysis techniques – Hardness determination by EDTA method, Alkalinity determination. Defects in boiler due to Hard water. External Treatment (Lime-soda, Zeolite & Ion exchange resin method) & Internal Treatment of Boiler feed water. Numerical Problems.
Unit - II	**FUELS & LUBRICANTS** **A. FUELS:** Characteristics of fuels. Classification of fuels, Calorific Value, HCV, NCV. Proximate and ultimate analysis of coal, Introduction to petroleum and its refining, knocking, octane number, & cetane number, Numerical problems. **B. LUBRICANTS:** Introduction, Classification &functions, Mechanism of lubrication, Properties of lubricating oils with significance: Viscosity Index, Flash point, Fire point, Aniline point, Cloud & pour point, Steam Emulsion Number (S.E.N), Numerical problems.
Unit – III	**CORROSION & ENERGY STORAGE SYSTEMS** **A. CORROSION:** Introduction, Types of Corrosion, Disadvantages of corrosion, Theories of corrosion, Factors influencing the rate of corrosion. Preventive measures of Corrosion like Galvanization, Tinning, Electroplating. **B. ENERGY STORAGE SYSTEMS:** Introduction, EMF of cell, Single electrode potential, Standard Hydrogen Electrode. Classification of batteries (primary, secondary and reserved batteries), Introduction to eV Batteries Advantages & Disadvantages. Construction, working, and applications of Li-ion batteries. Recycling of Lithium-ion batteries by direct cycling method.
Unit - IV	**POLYMERS & NANOMATERIALS** **A. POLYMERS:** Nomenclature & classification of polymers. Thermoplastics and Thermosetting polymers. Preparation, properties and applications of PE, PVC, PS, Teflon, Nylon 6:6, SBR, NBR, Bakelite, Silicone resin. Rubber & its type Vulcanization of Rubber, Photoactive polymers, solar cells and dye sensitized solar cells- principle and applications. Conducting Polymers: Methods of synthesis and properties of polyaniline (PANi), polypyrrol (PPy) and polythiophene (PTh). **B.NANOMATERIALS:** Synthesis, characterization and applications of nano-materials.
Unit – V	**INSTRUMENTAL METHODS OF ANALYSIS:** Importance of Instrumental techniques. Superiority of Instrumental Methods over conventional methods, Precision & Accuracy, Classification of Instrumental techniques. Introduction to Electromagnetic spectrum, Electroanalytical and Spectroscopic Methods. Principle, Instrumentation, Working and applications of following techniques: Colorimetry, IR Spectroscopy, Conductometry, pH metry, Chromatography and Gas Chromatography.

Textbooks:

- Engineering Chemistry – Jain & Jain – Dhanpat Rai &Company Pvt. Ltd, New Delhi.
- A Text Book of Engineering Chemistry – S.S. Dara – S. Chand Publication, Delhi.
- Engineering Chemistry- Shashi Chawla, Dhanpat Rai &Company Pvt. Ltd, Delhi.
- Engineering Chemistry – Uppal – Khanna Publishers.
- A Text book of Engg. Chemistry- Agarwal, C.V, Murthy C.P, Naidu, BS Publication, Hyderabad.
- B. Sivasankar, Engineering Chemistry 1 st Edition, Mc Graw Hill Education (India), 2008
- O.G. Palanna, McGraw Hill Education (India) Private Limited, 9 th Reprint, 2015

Reference Books:

- Chemistry in Engineering and Technology, Kuriacose J.C. and Rajaram J.,Tata McGraw Hill.
- Applied Chemistry- Theory and Practice, O.P. Viramani, A.K. Narula, New Age International Pvt. Ltd. Publishers, New Delhi.
- Chemistry of Engineering Material-C.V. Agarwal, Andranaidu C. Parameswara Moorthy –B.S. Publications.
- William Kemp, Organic Spectroscopy, 3 rd edition, Palgrave, New York, 2005.

List/Links of e-learning resource

- Advanced Materials and Processes https://onlinecourses-archive.nptel.ac.in/noc18_mm12/preview
- Lubricants & Lubrication https://nptel.ac.in/courses/112102015
- Cement: https://nptel.ac.in/courses/105104157
- Fuels: https://nptel.ac.in/courses/113104008

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Session	Batch 2026-27 onwards

Category of Course	Course Title	Course Code	L	T	P	Credits - 1
BSC	Applied Chemistry	CH26101P	-	-	2	

Course Code	Course Title	Maximum Marks Allotted					Total Marks
		Theory Paper			Practical		
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	
CH26101P	Applied Chemistry	--	--	--	30	20	50

Course Objectives	To develop practical skills in quantitative chemical analysis using classical volumetric methods and the characterization of water, fuels, and lubricants. The course provides hands-on training in instrumental techniques including colorimetry, chromatography, pH metry, and conductometry for qualitative and quantitative analysis.
Prerequisite Knowledge	Students should have basic knowledge of Chemistry, chemical reactions, solutions, concentration calculations, and fundamental analytical techniques. Familiarity with laboratory safety practices and basic titration methods will be helpful for performing the experiments.
Course Description	The course focuses on practical analytical chemistry through laboratory experiments involving volumetric analysis, water quality assessment, fuel and lubricant testing, and material characterization. Students gain hands-on experience with instrumental techniques including colorimetry, chromatography, pH metry, and conductometric analysis, developing essential skills in chemical measurement and data interpretation.
Course Outcomes	**Upon completion of the course, student will be able to (Please co-relate 05 Co's for 10 practical's)** **1.** Apply volumetric analysis techniques to determine the concentration and composition of chemical substances using acid-base, redox, complexometric, iodometric, and precipitation titrations. **2.** Perform quantitative analysis of water samples by evaluating hardness, alkalinity, chloride content, and other quality parameters using standard chemical methods. **3.** Analyze the properties of industrial materials such as coal and lubricating oils through proximate analysis, viscosity measurements, and flash/fire point determinations. **4.** Demonstrate hands-on proficiency in instrumental analytical techniques including colorimetry, pH metry, conductometry, and chromatography for qualitative and

	quantitative chemical analysis. **5.** Interpret experimental data, perform calculations, and evaluate analytical results with accuracy, precision, and adherence to laboratory safety and good laboratory practices (GLP).

List of Experiments: (Any 10 experiments to be performed)

Please list 10 experiments, if possible.

1.	To determine strength of unknown Ferrous Ammonium Sulphate $FeSO_4.(NH_4)_2SO_4.6H_2O$ (Mohr's Salt) solution by titrating it against intermediate Potassium Dichromate ($K_2Cr_2O_7$) solution using Di Phenyl Amine(DPA) as internal indicator.[Redox Titration]
2.	To determine Temporary, Permanent and Total Hardness in given sample of water by E.D.T.A. method.[Complexometric Titration]
3.	To determine strength of Sodium Carbonate and Sodium Bicarbonate in given alkaline solution by titrating with standard HCl using phenolphthalein and Methyl Orange indicators. Or To determine alkalinity in given water sample using Phenolphthalein and Methyl Orange indicators.[Acid Base Titration]
4.	To determine strength of unknown $CuSO_4$ solution by titrating it against intermediate sodium thiosulphate (Hypo) solution using starch as final indicator.[Iodometric Titration]
5.	To determine the chloride content of the given sample of water using silver nitrate solution with potassium chromate solution as an indicator.[Precipitation Titration]
6.	To determine Moisture content in given sample of coal.[Proximate Analysis]
7.	To determine Ash content in given sample of coal.[Proximate Analysis]
8.	To determine the Viscosity Index of give lubricating oil by Redwood Viscometer No.1 and Redwood Viscometer 2.[Lubricating Oil Analysis]
9.	To determine the Flash Point and Fire Point of lubricating oil by Abel's Apparatus.[Lubricating Oil Analysis]
10.	To determine the Flash Point and Fire Point of lubricating oil by Pensky Martin's Apparatus.[Lubricating Oil Analysis]

11.	To separate mixture of pigments by Paper Chromatography [Instrumental Methods].
12.	To verify Beer Lambert's law of colorimetry [Instrumental Methods].
13.	To determine amount of Iron by colorimetry [Instrumental Methods].
14.	To determine pH of given solution using pH meter. [Instrumental Methods]
15.	To determine strength of acid/base by conductometric titrations. [Instrumental Methods]
16.	Any other experiment set by department